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CAPSTONE PROJECT

FINAL CAPSTONE DISSERTATION

Final Dissertation Structure

Project Title	SmartDesk: A Customer Support Ticketing System for Small IT Support Teams
Group Members	Freek Schaefer-Gerber, Wian Sandenbergh
Module	SPM622 / MDB622
Lecturer / Supervisor	Mr. Dunstan
Week / Stage	Final Submission
Submission Date	29 October 2026

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Declaration

Declare originality and proper acknowledgement of sources.

We, Freek Schaefer-Gerber and Wian Sandenbergh, declare that this dissertation is our own original work. All sources used have been appropriately acknowledged and referenced. This work has not been submitted for assessment in any other module or institution. All demonstration data, user accounts, and support tickets used in the development and testing of the SmartDesk system are synthetic and do not contain any real personal or confidential information.

Abstract

Summarise the problem, aim, methodology, solution, key results and conclusion in one concise section.

Small IT support teams often rely on shared inboxes, spreadsheets, or informal messaging to manage customer requests, leading to lost tickets, unclear ownership, and poor visibility. This project details the design, development, and evaluation of SmartDesk, a web-based customer support ticketing system tailored for small IT support teams. The system aims to centralise requests into a secure, traceable workflow. Developed using Python, the Flask framework, and a SQLite relational database, SmartDesk implements role-based access control, secure password hashing, CSRF protection, and strict customer data isolation.

A lightweight Agile methodology was adopted for development. The system was evaluated using a combination of automated pytest suites and manual system, security, and usability tests. All automated tests passed, and twenty manual test cases confirmed that core requirements—such as authentication, validated ticket creation, staff-only assignment, and attributable comments—were met. Five defects were discovered and corrected during testing, including a critical cross-customer access flaw. The evaluation concludes that SmartDesk successfully addresses the core problem at a minimum viable product (MVP) level. While features such as audit trails and automated notifications remain out of scope, the project provides a secure and functional foundation for small teams.

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List of Abbreviations

Define abbreviations used in the report.

- **API:** Application Programming Interface
- **CSRF:** Cross-Site Request Forgery
- **CSS:** Cascading Style Sheets
- **HTML:** HyperText Markup Language
- **IT:** Information Technology
- **MVP:** Minimum Viable Product
- **NFR:** Non-Functional Requirement
- **RQ:** Research Question
- **SLA:** Service Level Agreement
- **SQL:** Structured Query Language
- **SRS:** Software Requirements Specification
- **UML:** Unified Modeling Language
- **URL:** Uniform Resource Locator
- **UX/UI:** User Experience / User Interface

Key Terms

Define important technical or research terms.

- **CSRF Token:** A security measure used to ensure that state-changing requests (like submitting a form) originate from the authenticated user and not a malicious third party.
- **Customer Isolation:** A security mechanism ensuring that a user can only view and interact with their own data (tickets) and cannot access the records of other users.
- **Hash (Password):** A one-way cryptographic transformation of a password. The system stores only the hash, meaning the original password cannot be recovered even if the database is compromised.

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- **Role-Based Access Control (RBAC):** A method of restricting system access to authorised users based on their specific role (e.g., Customer, Agent, Administrator).
- **Flask:** A lightweight web framework for Python used to build the server-side application layer.
- **SQLite:** A lightweight, file-based relational database management system used as the runtime database for the application.

Chapter 1 - Introduction

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Chapter 2 - Literature Review

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Chapter 3 - Requirements & Methodology

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Chapter 4 - System Design & Development

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Chapter 5 - Testing, Results & Evaluation

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Chapter 6 - Conclusion & Recommendations

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Appendices

Include supporting evidence such as questionnaires, extended test results, screenshots, user guide and selected project evidence.

- **Appendix A:** Software Requirements Specification (SRS)
- **Appendix B:** Database Schema and ERD (Paste the table and the image)
- **Appendix C:** Test Cases and Results (Paste the Excel table)
- **Appendix D:** Defect Log (Paste the Excel table)
- **Appendix E:** Requirements Traceability Matrix (Paste the Excel table)

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